FLIGHT PATTERNS

Some drone use permissions given in the past one year

- **Navi Mumbai Municipal Corporation** For surface topographical mapping of Navi Mumbai area
- **Asteria Aerospace** For survey, inspection and monitoring of Reliance pipelines
- **Indian Council of Medical Research** For vaccine delivery in Andaman & Nicobar Islands, Manipur and Nagaland
- **National Health Mission** Delivery of healthcare essentials in Jawhar tribal areas, Maharashtra
- **Karnataka govt** Creating urban property ownership records, Bengaluru
- **National Thermal Power Corporation** Conducting aerial surveillance, ash dyke monitoring and thermal inspection
- **Mahindra & Mahindra** Agricultural trials and precision spraying on paddy in Telangana and hot pepper crop in AP
- **Gangtok Smart City Development** *for aerial survey for Smart City Project*

(Source: Ministry of Civil Aviation)

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Mapping and surveys are currently the most lucrative commercial market for drones. They are used by mining companies and not too long ago Indian Oil deployed drones to monitor its 120 km Delhi-Panipat pipeline. "While a few government schemes are among the largest users of drones, the private sector is also picking up at a fast rate," says Shah. The Drone Federation of India estimates revenue potential of Rs 50,000 crore for the drone ecosystem in the next five years. Some of the early action is visible. In 2019, Reliance Industries Ltd picked up a strategic stake in Asteria Aerospace while in May this year Adani Defence Systems and Technologies Ltd acquired a 50 per cent stake in General Aeronautics, an agri-platform drone services provider. Both are Bengaluru-based companies.

Needless to say, cost and feasibility are going to dictate where drones are deployed. By that same token, getting a pizza delivered to your terrace isn't going to happen anytime soon. But it doesn't stop people from experimenting. Ankit Kumar of Skye Air says his firm is working with quick-commerce companies such as Swiggy to figure out ways to ferry orders in an urban setting, challenging as it is with wires, towers and buildings all in the way. "Flying in cities is 50 times more difficult than flying in rural or semi-urban environments," he says. The idea of hyperlocal delivery they are pursuing isn't so much about delivering to the customer's doorstep but ferrying groceries from a warehouse (dark store in industry parlance) to the nearest feasible point to a customer, so that the delivery boys can take over from there. Urban flying involves a detailed manual recce to mark out obstacles and garbage dumps where bird hits are possible, before venturing out. Over 80 per cent of Skye Air's 1,500 flights have been in urban centres like Gurugram, Noida and Bengaluru, he says. For logistics, Kumar reckons the early adopter is likely to be the medical diagnostics sector—with labs looking for a faster way to connect to their franchise offices spread out in various regions.

At the moment, the big thrust appears to be in agriculture. Here too, cost can be a crucial factor. With the subsidy, a farmer producer organisation (FPO) can now buy a Rs 10 lakh drone for about Rs 6 lakh and avail of finance options too. "Drone technology has proved to be viable in other Asian countries with smallholder farms," says Asitava Sen, CEO, CropLife India, an association of agro-tech firms. Several of them have approached the regulator for product approvals while trials are being held by start-ups, agriculture universities and FPOs, he says. "It's the perfect time as the policy framework is in place."

Deepak Bhardwaj of IoTechWorld explains that drones offer greater control and speed in crop-spraying compared to manual methods. The drones typically save water because the droplet size can be adjusted and the spray is targeted towards the canopy or stems of the plants thereby increasing the efficacy. Widespread adoption of kisan drones will depend on creating a network of service providers plus awareness through Krishi Vigyan Kendras, he says. "First let them get a taste of the technology, see the results and then they will come in," he says. Since June 14 when his agri-drone received type certification, IoTechWorld has generated about 100 unique identification numbers (UINs), he adds. The UIN is to a drone what a registration number is to a car with the registrations done on the DigitalSky platform.

The drone ecosystem in India is still in its early phase but growing. From a policy perspective, things are on par with what's happening globally, says Shah of DFI. "With the experiments happening for BVLOS (beyond visual line of sight) and air space integration, we will soon catch up aggressively on newer technologies." Drones, he says, are here to stay. ■